

EDGE AI ON RESOURCE-CONSTRAINED SENSOR NODES: Distributed Visual Monitoring For Scientific And Medical Instrumentation Approaches

Tsegay Teklay Gebrelibanos^{*1}, Carlo Vallati¹, Fabrizio Palla², Bruno Augusto Casu Pereira De Sousa², Piero Giorgio Verdini², Angela Zinnai³

¹Università di Pisa, Dipartimento di Ingegneria dell'Informazione, Largo Lucio Lazzarino 2, 56126, Pisa;

²Istituto Nazionale Fisica Nucleare INFN, Polo Fibonacci Largo B. Pontecorvo, 3, 56127, Pisa;

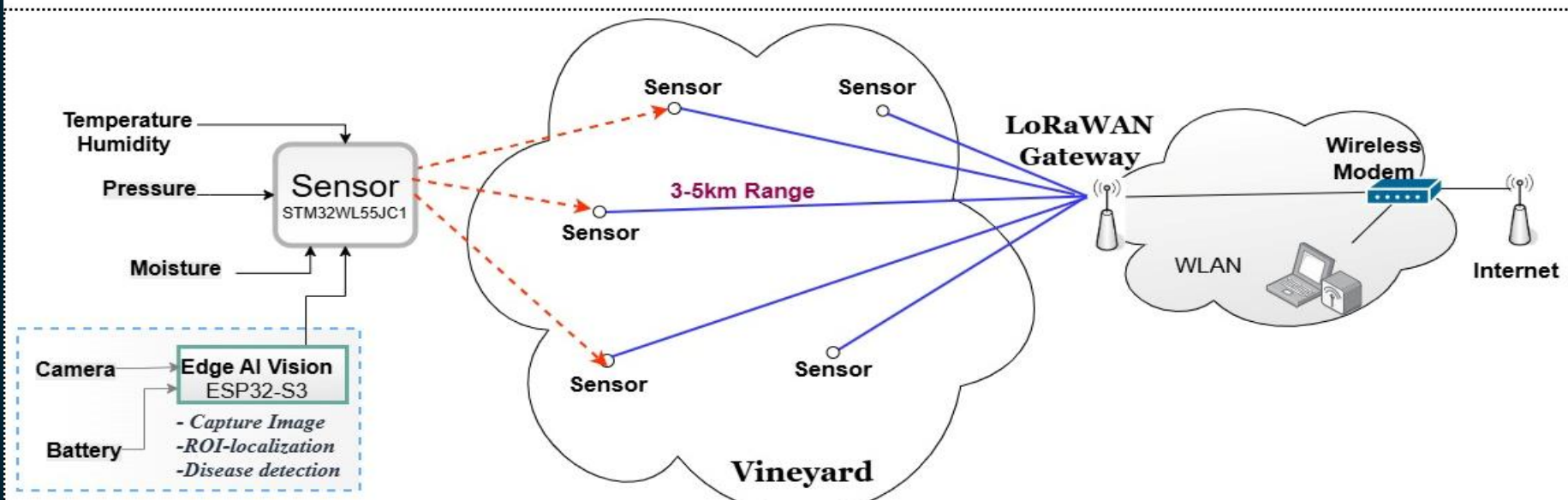
³Università di Pisa, Dipartimento di Scienze Agrarie, Alimentari e Agro-ambientali DISAAA-a, Via del Borghetto 80, 56124, Pisa;

INTRODUCTION

Distributed monitoring systems often rely on low-power wireless sensor networks that collect scalar environmental parameters (e.g., **temperature, humidity, pressure**), providing only indirect indications of system conditions. This work explores the integration of **Edge AI-based visual analysis** directly within resource-constrained sensor nodes (operating in a **LoRaWAN network**), enabling real-time condition monitoring while minimizing bandwidth, power consumption, and computational requirements.

The system uses a four-stage inference pipeline deployed on an **ESP32-S3** microcontroller with an **OV3660** camera. Operating in a duty-cycled mode with deep sleep between acquisitions, the system achieves an estimated 10-month lifetime on a 2400 mAh battery.

LoRaWAN monitoring system architecture with custom design End node (sensor device)



OBJECTIVES

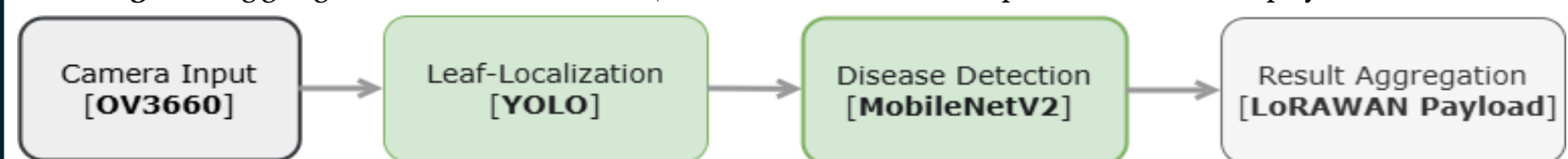
- Apply **visual-based Edge-AI detection** to confirm the presence of disease.
- Implement and deploy a complete inference pipeline on an ESP32-S3 microcontroller.
- Optimize leaf localizer and disease classifier models to fit the ESP32-S3 (8 MB PSRAM, 16 MB flash).
- Aggregate inference output of multiple leaves into single frame-level inference result (LoRaWAN payload).
- Analyse deployment trade-off: accuracy, latency, memory utilization, battery lifetime.



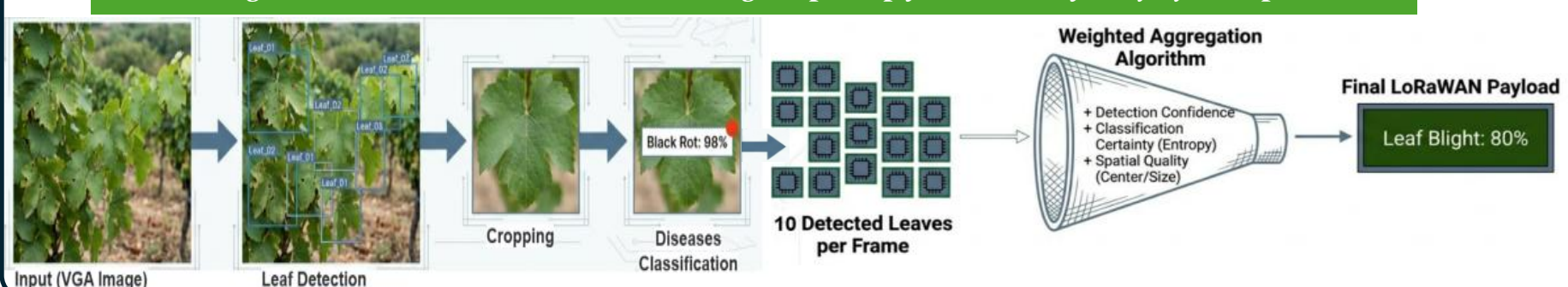
METHODOLOGY

The Four stage Inference pipeline

- Stage 1:** Input vine image captured using the OV3660 camera module embedded with ESP32.
- Stage 2:** Leaf Localization: a lightweight model to detect leaf region (YOLO).
- Stage 3:** Disease classification: classify each leaf into (Black Rot, Esca, Healthy, Leaf Blight).
- Stage 4:** Aggregate inference results; transmit in one compact LoRaWAN payload.

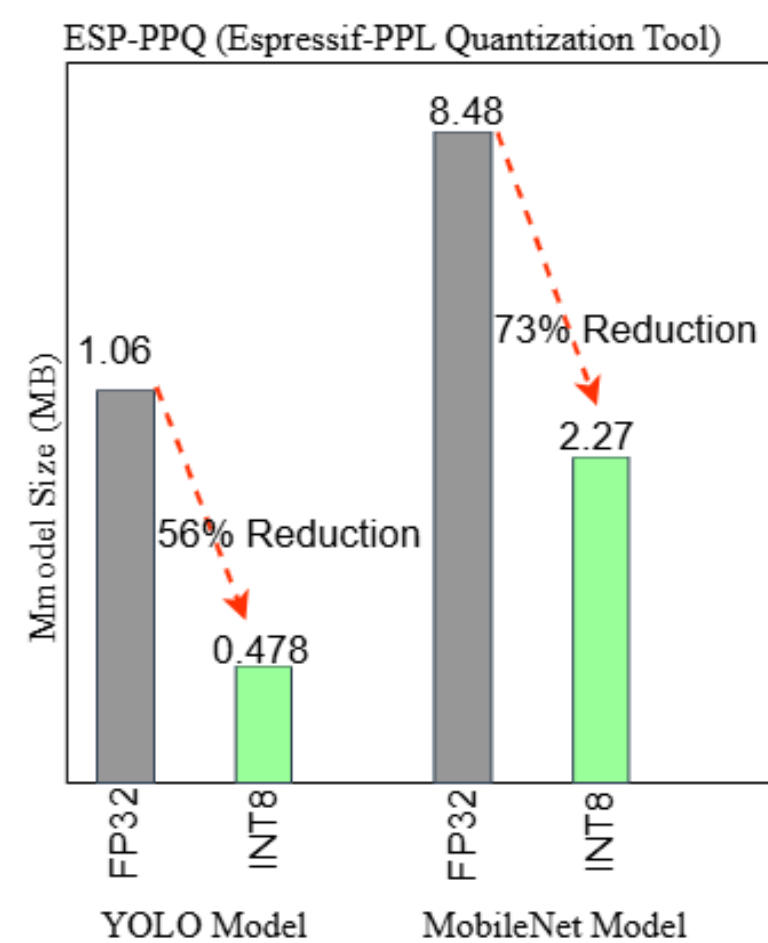


All the stages run on-ESP32-S3 device (entering deep sleep for 6-hour of duty cycles operation)

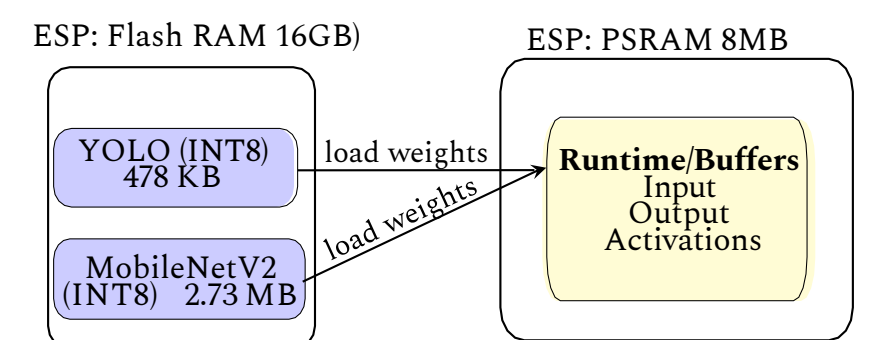


RESULTS AND PERFORMANCE EVALUATION

Following YOLO11n and MobileNetV2 model training and **INT8 quantization**, models stored in the 16MB flash RAM and only loaded into PSRAM during runtime for inference execution.

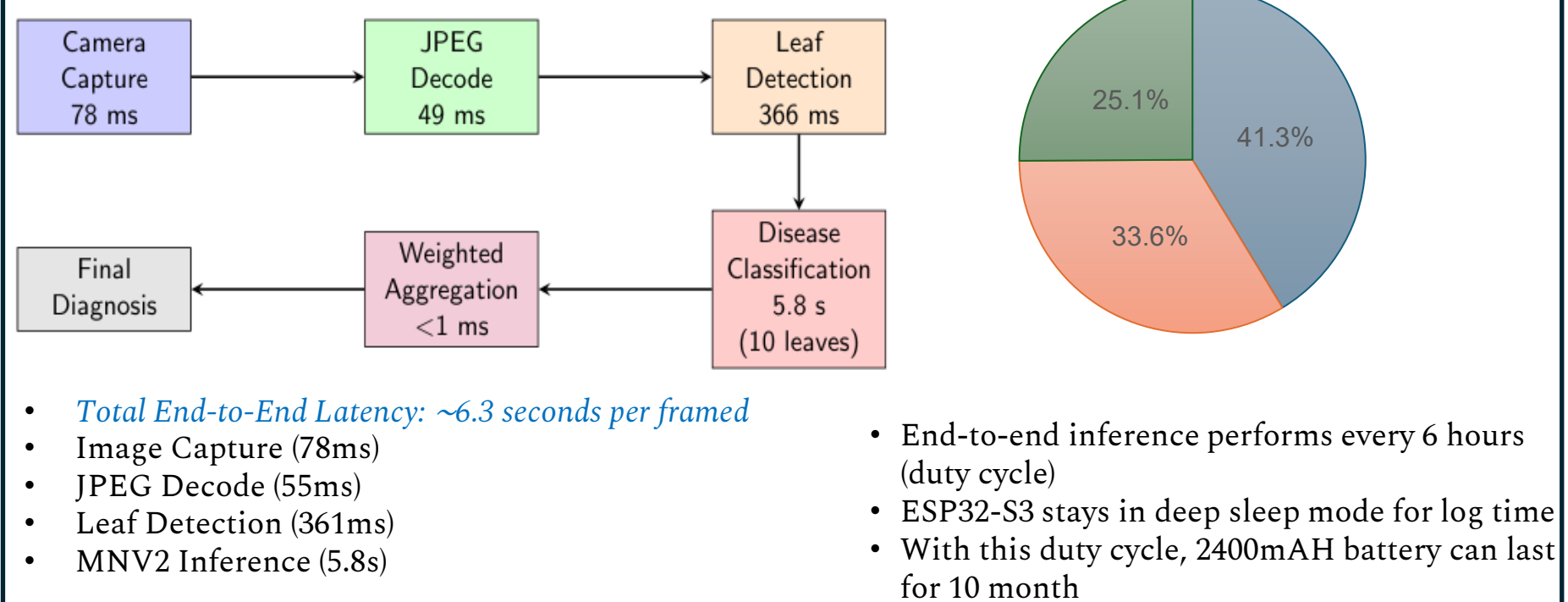


Metric	YOLO11n	MobileNet V2
Accuracy / mAP@50	56.4%	99.8%
Precision/F1-Score	64.8%	0.998
Parameters	0.36M	3.5M
Model Size (INT8)	478KB	2.27MB



Operational Phase Power Consumption

■ Inference Phase ■ Deep Sleep Phase ■ System Overhead



- Total End-to-End Latency: ~6.3 seconds per framed**
- Image Capture (78ms)
- JPEG Decode (55ms)
- Leaf Detection (361ms)
- MNV2 Inference (5.8s)
- End-to-end inference performs every 6 hours (duty cycle)
- ESP32-S3 stays in deep sleep mode for log time
- With this duty cycle, 2400mAh battery can last for 10 month

APPLICATION AREAS BEYOND AGRICULTURE: SCIENTIFIC & MEDICAL INSTRUMENTATION

Beyond the agricultural use case, the proposed architecture provides a general framework for embedded visual monitoring in scientific infrastructures.

HEP experimental facilities

Mobile robotic inspection, leaks, cable damage, structural anomalies

Medical Physics

Radiotherapy QA, medical imaging instrumentation, and phantom-based dosimetry

Remote Instrumentation

Distributed event-driven visual monitoring of harsh environments without continuous image transmitting

CONCLUSION

- Demonstrated fully embedded visual AI that runs on a resource-constrained sensor node.
- Validate **end-to-end inference pipeline** (Capture → Detect → Crop → Classify) on ESP32-S3.
- Analysis **accuracy, memory, latency, and battery lifetime** for real deployments.
- This grape-leaf disease detection case study demonstrates general framework to distributed visual monitoring for scientific (e.g., FCC-ee) and medical instrumentations.

REFERENCES

- Xu Tao, Jackson Butcher, Simone Silvestri, and Sajal K Das. icrop+: An edgeboosted crop disease detection system via tinyml and lora communication. In 2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops), pages 558–560. IEEE Computer Society, 2025.
- Zuxing Chen, Junjie Feng, Kun Zhu, Zhenyan Yang, Yanhong Wang, and Mingyue Ren. Yolo-v8-acw: Lightweight grape leaf disease detection method based on improved yolo-v8. IEEE Access, 12:123595–123608, 2024.
- Midhun P Mathew, Sajmi Salam, Sophiya Mathews, et al. Real-time automatic detection of grape leaf diseases with mobilenetv2 deep learning network based on artificial intelligence. In 2025 Advanced Computing and Communication Technologies for High Performance Applications (ACCTHPA), pages 1–6. IEEE, 2025